

The background of the slide is a collage of Lyft and Uber logos and images of cars. The Lyft logo, in its signature magenta color, is repeated several times in different sizes and orientations. The Uber logo, in blue, is also visible. Images of various cars, including a blue sedan, a green Dodge muscle car, and a red pickup truck, are scattered throughout the background, some appearing to be in motion or parked. The overall aesthetic is modern and tech-oriented.

Rideshare Price Prediction in Boston using Machine Learning

CIS 508: Final Project

Meshach Samuel

Business Problem & Stakeholders

Business Context	Stakeholders	ML Task
• Rideshare prices fluctuate with distance, demand and weather, which makes it hard for riders to anticipate cost	• Riders (Plan Trips, Compare Prices) • Drivers (Understand when Trips are More Profitable) • Platform (Pricing Transparency)	• Target = Price • Supervised Regression

Given a Rider's Pickup Details, Can we Predict the Ride Price?

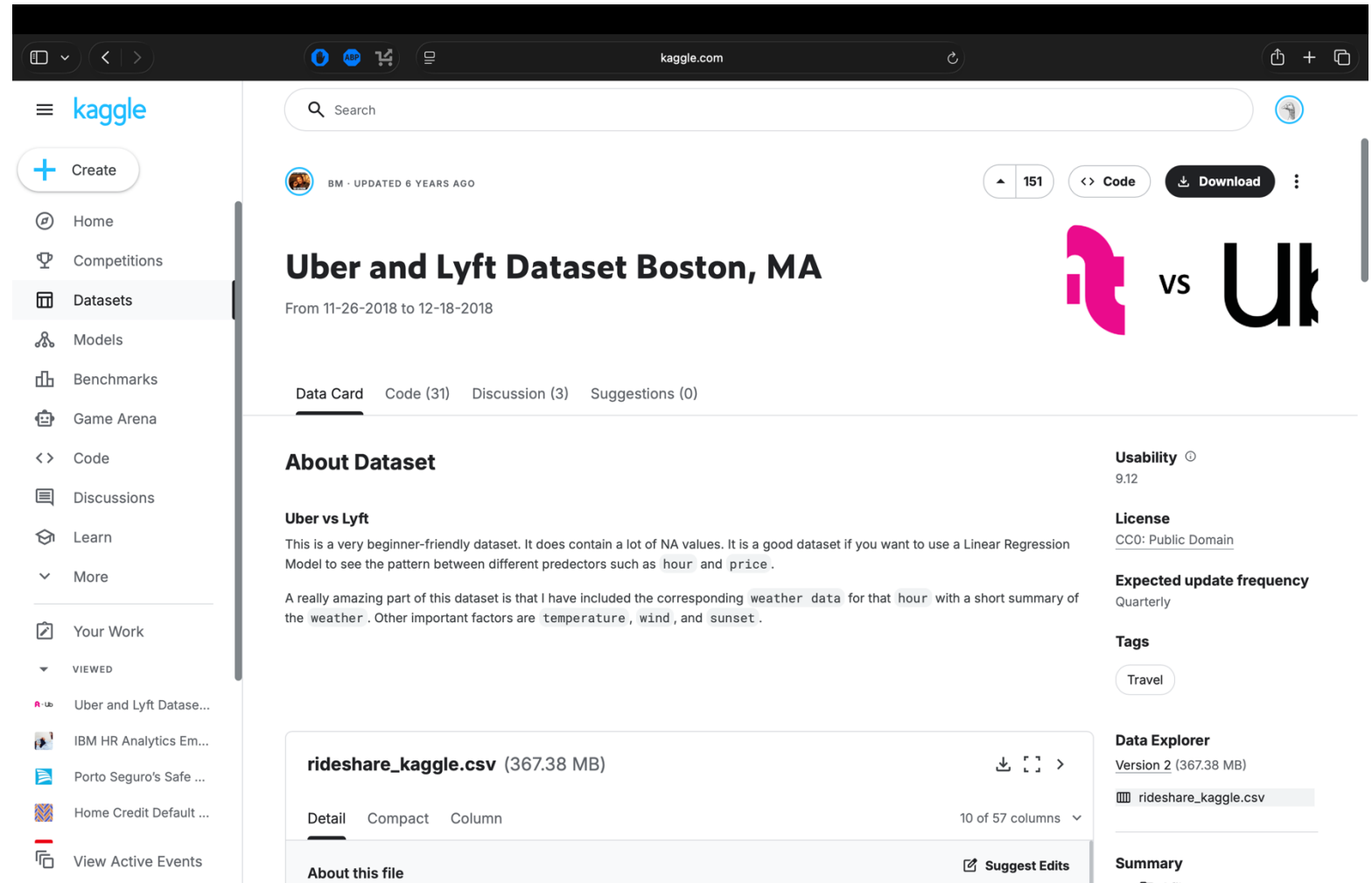
Dataset Overview

Source: Kaggle “Uber and Lyft Dataset Boston, MA”

693071 Rows, **57** Columns

Target: price

Key Features: hour, day, month, distance, name, cab_type, source, destination, etc.

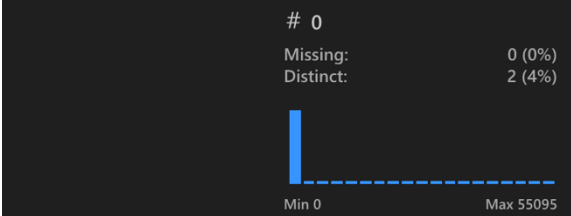


The screenshot shows the Kaggle website interface. On the left is a sidebar with navigation links: Home, Competitions, Datasets (highlighted), Models, Benchmarks, Game Arena, Code, Discussions, Learn, and More. The main content area displays the 'Uber and Lyft Dataset Boston, MA' page. At the top, there's a search bar and a 'Download' button. Below the title, there's a section 'About Dataset' with a description of the dataset, its features, and a 'Usability' score. The 'Data Explorer' section shows the dataset file 'rideshare_kaggle.csv' (367.38 MB) with a 'Detail' tab selected. The 'Summary' section is partially visible at the bottom.

<https://www.kaggle.com/datasets/brllrb/uber-and-lyft-dataset-boston-ma>

Data Quality and Checks

- Loaded Data with Pandas and inspected Summary Statistics
 - Missing Values – 55K
 - Duplicate Rows - None
 - Removed Unnecessary Columns: id, icon, timezone, etc.
 - Removed Outliers
- Only Price Column had Null Values
- 55K Rows with Null Values from 690K Rows. $55/690 = < 8\%$
Easiest to Remove All Rows with Null Values



0
Missing: 0 (0%)
Distinct: 2 (4%)

price	55095
id	0
ozone	0
temperatureLowTime	0
apparentTemperatureHigh	0
apparentTemperatureHighTime	0
apparentTemperatureLow	0
apparentTemperatureLowTime	0
icon	0
dewPoint	0
pressure	0
windBearing	0
cloudCover	0
uvIndex	0
visibility.1	0
sunriseTime	0
temperatureHighTime	0
sunsetTime	0
moonPhase	0
precipIntensityMax	0
uvIndexTime	0
temperatureMin	0
temperatureMinTime	0
temperatureMax	0
temperatureMaxTime	0
apparentTemperatureMin	0
apparentTemperatureMinTime	0
apparentTemperatureMax	0
temperatureLow	0
temperatureHigh	0

Final shape after outlier removal: (625229, 36)

Feature Selection

VIF Analysis:

- Used Variance Inflation Factor (VIF) to Identify Highly Collinear Numeric Columns
- Dropped Numeric Columns with very High Correlation: $VIF > 10$

Recomputing VIF with final numeric feature set:

	feature	# vif
0	windBearing	9.511397261373537
1	moonPhase	7.572177427786643
2	cloudCover	5.321989587782691
3	precipProbability	4.8549149339037045
4	windGust	4.327367922258211
5	distance	4.237078479906154
6	hour	4.2355297818329545
7	precipIntensity	4.214754635707543
8	precipIntensityMax	2.8004470435407702
9	uvIndex	1.4579996637836263

10 rows x 2 cols 10 per page

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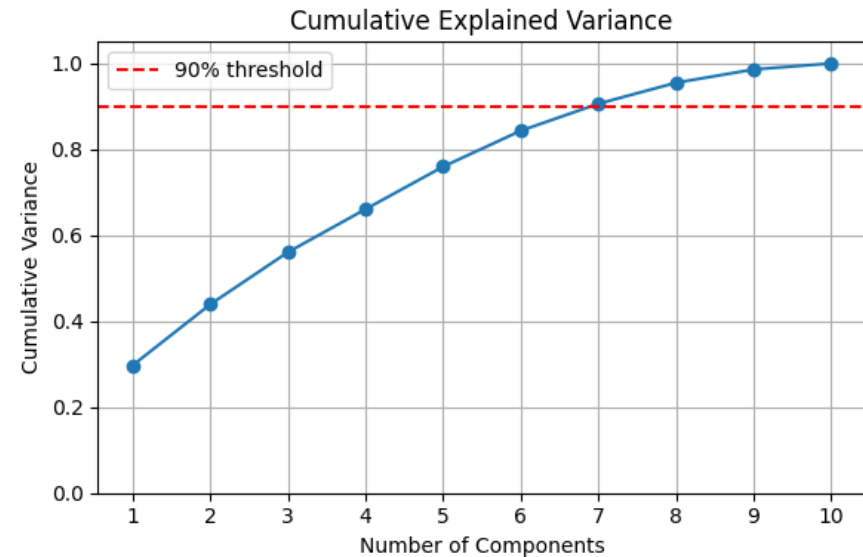
Dropped features due to high VIF:

- temperatureHigh: VIF = 2182290.7596
- temperature: VIF = 20975.1747
- apparentTemperatureMax: VIF = 4880.5560
- pressure: VIF = 2091.3553
- temperatureMax: VIF = 1512.0292
- temperatureMin: VIF = 957.6175
- dewPoint: VIF = 754.4140
- temperatureLow: VIF = 702.6919
- apparentTemperatureHigh: VIF = 178.1387
- ozone: VIF = 159.0812
- surge_multiplier: VIF = 93.9382
- apparentTemperatureMin: VIF = 88.6238
- windSpeed: VIF = 56.0298
- humidity: VIF = 50.1658
- apparentTemperature: VIF = 30.2647
- visibility: VIF = 20.2376
- apparentTemperatureLow: VIF = 11.9246

Feature Selection

PCA Analysis:

- Performed PCA on standardized numeric columns to understand feature redundancy
- First few components captured most of the variance
- Dropped Columns after exceeding 90% Variance



Number of PCs required to reach 90 percent variance: 7

Shape of PCA transformed data: (625229, 7)

Feature importance based on average absolute PCA loadings:

# 0	
cloudCover	0.29371921588927064
windGust	0.29354862878814697
uvIndex	0.28938279262387756
moonPhase	0.2880113035570671
hour	0.2826651783340682
windBearing	0.19616836748413455
distance	0.18090595342000815
precipIntensit	0.1717457039351626
precipIntensit	0.16862031053335833
precipProbabi	0.13115474397036267

Final Feature Selection List

Numeric List:

- hour
- distance
- precipIntensity
- windGust
- windBearing
- cloudCover
- uvIndex
- moonPhase
- precipIntensityMax

Categorical List:

- cab_type
- name
- source
- destination
- short_summary
- day_name
- month_name
- is_weekend
- is_peak_hour

Modelling Pipeline

- **Input:** Full feature matrix with numeric and categorical columns
- **Pre-Processing:**
 - Numeric Pipeline: StandardScaler for All Numeric Columns
 - Categorical Pipeline: OneHotEncoder
- Each Model wrapped in a Pipeline with the Pre-Processor
- Train-Test Split: 80-20 Split with Random State

Models: Algorithms and Tuning

Models Trained:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
- KNN Regressor
- Linear SVM
- Neural Network
- Ensemble Model - Gradient Boosting

Hyper-Parameter Tuning

- Used GridSearchCV
- Each Model had 3 Parameters with 2 Options

Scoring Metric

- RMSE (Root Mean Squared Error)

Model Comparison

Model comparison (sorted by RMSE):

	model	best_params	# rmse	# mae	# r2
0	XGBRegressor	{'model__learning_rate': 0.1, 'model__	1.9869716973561327	1.2316164577971866	0.9480133293514258
1	MLPRegressor	{'model__alpha': 0.001, 'model__hidd	2.0029401455430764	1.24411897380737	0.9471743820980391
2	RandomForestRegressor	{'model__max_depth': 15, 'model__m	2.0128105385047683	1.2455039576558338	0.9466524550209205
3	DecisionTreeRegressor	{'model__max_depth': 20, 'model__m	2.065186147604914	1.272361714244715	0.9438400063944092
4	GradientBoostingRegressor	{'model__learning_rate': 0.1, 'model__	2.123672831825457	1.36970242595578	0.9406140285269383
5	LinearRegression	{'model__fit_intercept': True, 'model__	2.45466371639252	1.7335950234279849	0.9206599172143236
6	LinearSVR	{'model__C': 10.0, 'model__epsilon': 0	2.5042293420157002	1.6889403400365763	0.9174234293876246
7	KNeighborsRegressor	{'n_neighbors': 10, 'weights': 'distan	3.3634759125036653	2.1268623747772963	0.8510346413014092

8 rows x 5 cols

10 ▾ per page

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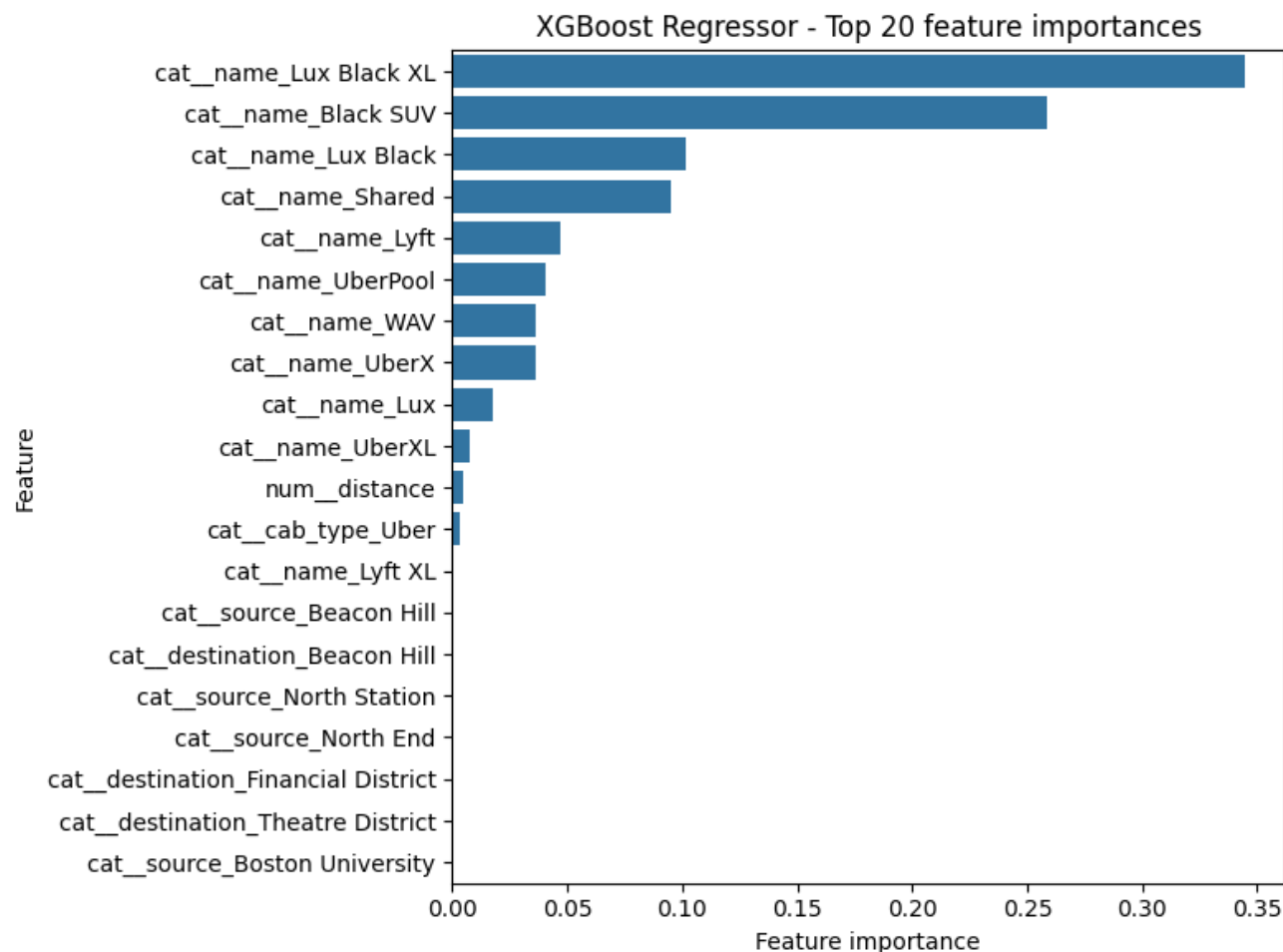
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Best model based on RMSE: XGBRegressor

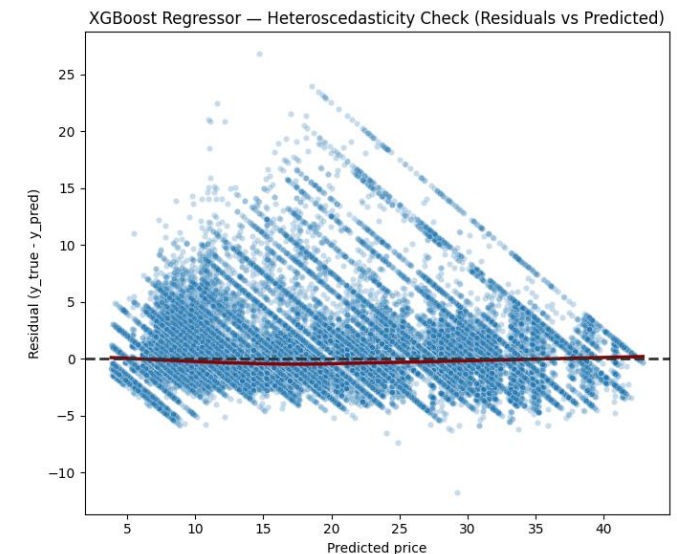
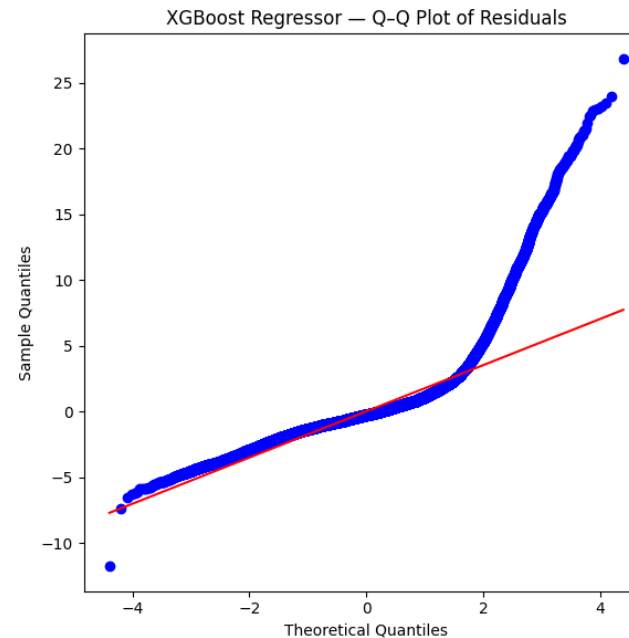
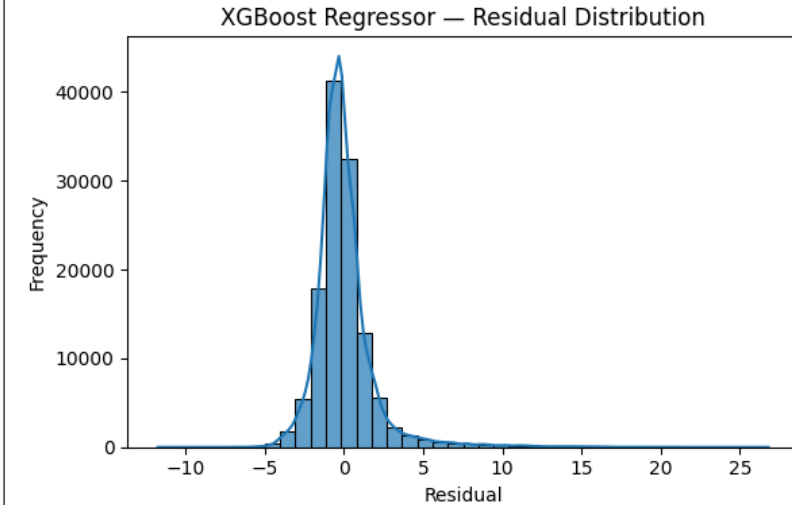
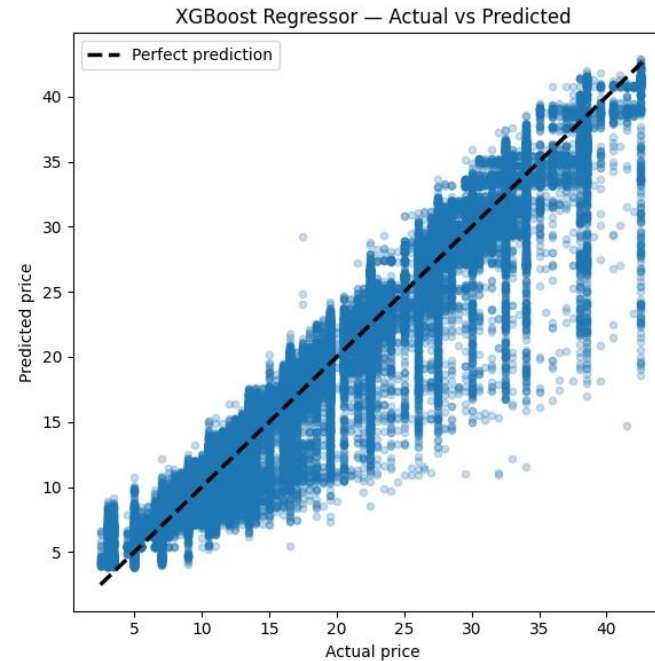
Feature Importance

- Majority of the Important Features are **Cab Type**
- **Distance** is also an Important Feature
- **Origin** and **Destination** of Ride are also Important Features



Error Analysis

- Predictions closely follow the perfect fit line, with larger spread on higher priced rides
- Residuals center around zero with a slight positive tail showing occasional underprediction
- Residuals match normality in the middle but deviate in the upper tail, confirming heavier positive errors
- Residual variance increases with predicted price, indicating mild heteroscedasticity



Model Deployment

<https://meshachsamuel-cis508-finalproject.streamlit.app>

Limitations and Future Improvements

Limitations:

- **Data:**
 - Single City (Boston) and Short Period
 - No Traffic or Real Time Data
- **Modeling:**
 - Pure Supervised Regression. No Time Series Structure
 - No Analysis by Neighbourhood

Improvements:

- **Integrate Additional Features:**
 - Real Time Demand
 - Traffic Conditions
 - Calendar Events
- Extend to Multiple Cities and Compare Pricing Patterns across Cities and Locations

Links

- **GitHub:** <https://github.com/ProntoSbinalla/CIS-508-Final-Project>
- **DataBricks:** https://dbc-c0894453-af7b.cloud.databricks.com/ml/experiments/927873117134361/runs?o=3434810423377078&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Active&modelVersionFilter=All+Runs&datasetsFilter=W10%3D
- **StreamLit App:** <https://meshachsamuel-cis508-finalproject.streamlit.app>